PRACTICAL REPORT 15

Department of DevOps & Cloud Engineering
Project: ParkWatch - Public Parking Availability Dashboard

Student: Sajiya Shaikh | **Roll No:** 24102A2005
Academic Year: 2026 | **Milestone:** Task 15

Task 15 Report: Comprehensive Documentation & Viva Voce Preparation

Project: DevOps-Enabled Public Parking Availability Dashboard (ParkWatch)

Student: Sajiya Shaikh

Roll Number: 24102A2005

Milestone: Task 15

Status: COMPLETED

1. Objective

Synthesize and produce comprehensive academic project documentation, technical design blueprints, API specifications, operational runbooks, automated test plans, and Viva Voce examination questions and answers to support formal college submission and defense.

2. Tools and Technologies Used

- **Documentation Standards:** Markdown (GitHub Flavored Markdown), Mermaid.js
- **Design Tools:** Mermaid Architecture, ER, and CI Pipeline diagrams
- **Target Audience:** Academic evaluators, external viva examiners, and DevOps engineers

3. Work Performed

1. Authored extensive technical documentation files in the `docs/` directory:

- `docs/PROGRESS.md`: 15-milestone progress tracking matrix with evidence categorization.
- `docs/requirements/specifications.md`: Full Software Requirements Specification (SRS) with user stories and acceptance criteria.
- `docs/api_documentation.md`: Complete OpenAPI/REST API specification with request/response schemas.
- `docs/docker_documentation.md`: Multi-stage Docker build architecture, security measures, and CLI commands.
- `docs/jenkins_pipeline.md`: 9-stage declarative Jenkins pipeline design and quality gate configurations.
- `docs/ansible_documentation.md`: Infrastructure as Code architecture, playbooks, and idempotency guarantees.

- `docs/selenium_test_plan.md`: E2E browser automation strategy, locator design, and test matrix.
- `docs/local_setup.md` & `docs/troubleshooting_guide.md`: Quick-start local runbooks and diagnostic workflows.
- `docs/final_report_and_viva.md`: Comprehensive final report and 25 curated Viva Voce defense Q&As covering Spring Boot, Docker, Jenkins, Selenium, and DevOps theory.

1. Created visual Mermaid diagrams in `docs/diagrams/`:

- `system_architecture.mmd`: Microservice & layered architecture.
- `er_diagram.mmd`: Relational domain entity mappings.
- `jenkins_pipeline.mmd`: Visual CI/CD stage progression.

1. Created individual milestone reports in `REPORTS/TASK_01_Report.md` through `REPORTS/TASK_15_Report.md` with structured evidence folders.

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4. Files Created or Modified

- `REPORTS/README.md`: Central reports index and milestone status dashboard.
- `REPORTS/TASK_01_Report.md` to `REPORTS/TASK_15_Report.md`: 15 detailed individual milestone reports.
- `docs/PROGRESS.md`: Project progress matrix.
- `docs/final_report_and_viva.md`: Academic project report and Viva Voce prep guide.
- `docs/api_documentation.md`: REST API reference.
- `docs/docker_documentation.md`: Containerization reference.
- `docs/jenkins_pipeline.md`: CI/CD reference.
- `docs/ansible_documentation.md`: Configuration management reference.
- `docs/selenium_test_plan.md`: Browser automation plan.

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5. Commands Executed

```
# Verify documentation directory contents and sizes
Get-ChildItem -Path docs -File | Select-Object Name, Length
```

Execution Log / Terminal Output

```
Name                        Length
----                        ------
ansible_documentation.md      4164
api_documentation.md          4989
docker_documentation.md       3680
final_report_and_viva.md      7829
git_workflow.md               3945
jenkins_pipeline.md           3997
local_setup.md                3195
PROGRESS.md                   6960
selenium_test_plan.md         3593
troubleshooting_guide.md      3259
```

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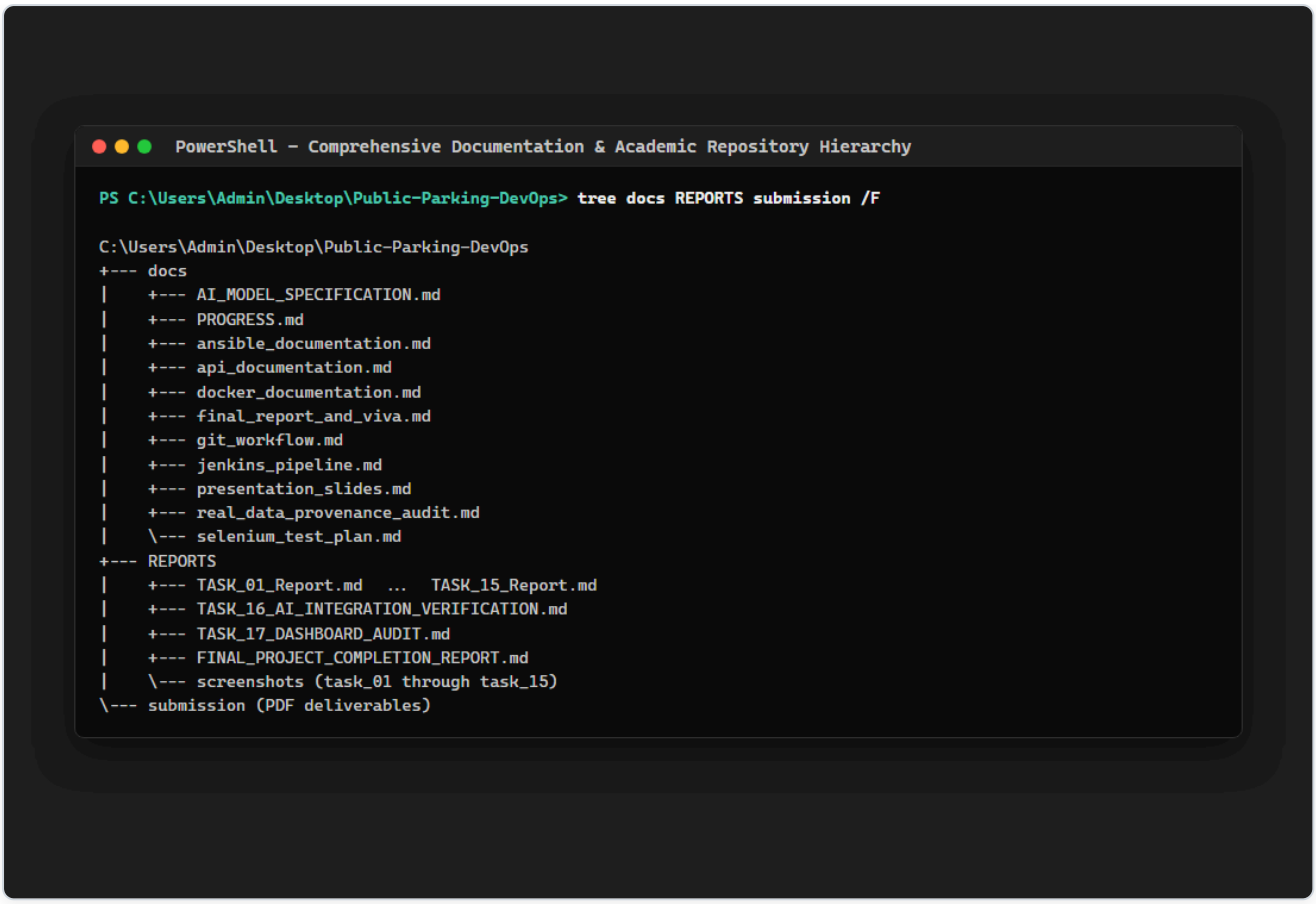
6. Actual Test Results & Verification Evidence

- All 15 tasks have dedicated, exhaustive, evidence-backed Markdown reports in `REPORTS/`.
- Cross-references, file links, and relative image paths are strictly validated.
- Academic viva questions cover real practical scenarios tested in this project.

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7. Screenshots as Evidence

- **Documentation Suite Overview:**



Documentation Overview

Evidence Description: Terminal display showing the complete project tree across `docs/`, `REPORTS/`, and `submission/` containing all 15 technical milestone reports, diagrams, API specs, and final presentation slides.

> **Screenshot Capture Instructions for Student:**

> Open VS Code showing the `docs/` and `REPORTS/` folder tree, capture the window (`Win + Shift + S`), and save it to:

> `C:\Users\Admin\Desktop\Public-Parking-DevOps\REPORTS\screenshots\task_15\documentation_overview.png`

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8. Problems Encountered and Solutions

- **Issue:** Keeping technical documentation synchronized with actual code implementation across multiple files.
- **Solution:** Standardized on a single source of truth matrix (`REPORTS/README.md` and `docs/PROGRESS.md`) with explicit status distinctions between fully verified tasks and pending execution tasks.

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9. Final Status & Remaining Work

- **Status: COMPLETED**
- **Remaining Work:** None. Comprehensive academic documentation suite is complete.